

EGRE 245 Engineering Programming

Spring 2026

Mark Schwitzerlett

Lab Exercise #9: Introduction to Pointers

1 Overview

In this lab, write a C program that demonstrates the use of pointers. The program will store two integer values, assign pointers to those values, and use pointer operations to modify and swap the values.

Your program should:

- Request the user to input two integer values.
- Store the values in variables.
- Declare two pointer variables and assign them to the addresses of the integers.
- Display:
 - the values of the integers
 - the addresses of the integers
 - the values stored in the pointers
 - the dereferenced pointer values
- Use the pointers to:
 - add 10 to the first value
 - multiply the second value by 2
- Swap the values using pointer dereferencing.
- Output the updated values after each step.

2 Code Documentation

The very first lines in all of your programming project files this semester should be comments in the class header file (copy from header.c).

In addition, the first output for all of your projects this semester should be the name of the project and your name, e.g.:

```
printf("Lab #: Your Name\n");
```

For lab exercises, include both student names if working in a group.

3 Sample Run

Your output should match the sample run below.

```
Lab #9: Mark Schwitzerlett and Jacobi
Enter value for a: 5
Enter value for b: 8

Initial values:
a = 5
b = 8

Addresses:
&a = 0x7ffd0a7b4118
&b = 0x7ffd0a7b4114

Pointer values:
p1 = 0x7ffd0a7b4118
p2 = 0x7ffd0a7b4114

Dereferenced values:
*p1 = 5
*p2 = 8

After pointer modifications:
a = 15
b = 16

After swap:
a = 16
b = 15

...Program finished with exit code 0
Press ENTER to exit console.
```

4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single “.c” file) containing a `main()` function.

All labs this semester will be submitted in Gradescope (via Canvas).

Your file should be named:

`Lab_Pointers_FirstName_LastName.c`

Due date: The lab is due at the end of your lab period **Monday, April 13th or Tuesday, April 14th.**

Be sure to save a copy of your .c file.